Clinical profiles and free-living glucose in type 2 diabetes: a bidirectional analysis in the AI-READI cohort

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Draft, October 19, 2026

Abstract

Background. Discrete clinical subtypes have been proposed to guide treatment selection in type 2 diabetes. However, analyses of clinical trial data show that the underlying continuous clinical variables can predict disease progression and treatment response as well as or better than subtype labels. How closely these subtypes correspond to distinct patterns of free-living glucose remains unclear.

Methods. In the AI-READI cohort, we derived three clinical profiles (C1, C2, C3) from six clinical variables among 458 participants with type 2 diabetes not treated with insulin. We compared mean glucose, time in range and coefficient of variation across profiles from continuous glucose monitoring (CGM). We then reversed the direction: we grouped participants from CGM features alone and tested whether these groups matched the profiles and differed in clinical variables. We repeated the analysis across all glycemic strata (1,591 participants).

Results. Clinical profiles captured modest differences in glucose behavior, with small-to-medium effects across the three CGM outcomes (η^2 0.042–0.078) and substantial overlap between profiles. Grouping participants by CGM instead revealed similarly limited correspondence with clinical profiles (ARI 0.039). These CGM groups differed mainly in HbA1c, with smaller differences in the other clinical variables (η^2 0.002–0.061). Consistent with this pattern, CGM features predicted HbA1c ($R^2 = 0.59$) but showed negligible predictive value for C-peptide and triglyceride-to-HDL ratio. CGM features also predicted profile membership only modestly (macro AUROC 0.649), while their addition to HbA1c increased AUROC from 0.605 to 0.660. Correspondence was weaker in healthy and prediabetes participants, whose clinical profiles differed only in mean glucose and were poorly recovered by CGM grouping (ARI $\leq$ 0.013). Applying the T2D profiles across strata showed that the C2-associated reduction in variability was approximately 40% as large in prediabetes and absent in healthy participants. Across the pooled cohort, CGM groups aligned more closely with diagnostic strata (ARI 0.160), indicating that broad differences across the glycemic spectrum exceeded the correspondence observed between profiles within type 2 diabetes. Across two non-overlapping 3-day windows, CGM groups were moderately reproducible (ARI 0.42), while their agreement with the profiles stayed low in each window (ARI 0.030 and 0.036).

Conclusions. Clinical profiles captured modest differences in free-living glucose but showed limited agreement with CGM-derived groups. HbA1c was the strongest link between clinical characteristics and glucose patterns, although CGM provided additional information about profile membership. Clinical and CGM-based groupings therefore describe related but distinct aspects of glycemic heterogeneity.

1 Introduction

Type 2 diabetes is heterogeneous in its progression, complication risk and treatment response. This heterogeneity has motivated clinical subtypes to support risk stratification and treatment selection. Ahlqvist et al. identified five subgroups from age at diagnosis, BMI, HbA1c, HOMA2-β, HOMA2-IR and GAD antibody status [1].

Analyses of the ADOPT and RECORD trials found that continuous clinical variables predicted disease progression as well as or better than cluster membership [4]. Similarly, a comparison using TriMaster trial data found greater estimated HbA1c benefit from treatment allocation based on continuous features than on clinical clusters [6]. However, the ORIGIN trial identified subtype differences in renal outcomes and response to insulin glargine [VERIFY: ORIGIN citation]. These findings suggest that clinical subgroups may capture relevant heterogeneity without consistently providing more useful predictions than their underlying continuous variables.

Clinical subtyping and CGM phenotyping describe heterogeneity from different perspectives. Clinical profiles group individuals by metabolic and demographic characteristics, whereas CGM-defined groups summarize their observed glucose patterns [VERIFY: Hall et al. 2018, PLoS Biology]. Previous CGM clustering studies have identified groups with differing clinical characteristics, but such differences do not establish that glucose-based and clinically defined partitions identify the same individuals. Evaluating their correspondence requires examining both glucose outcomes across clinical profiles and clinical characteristics across CGM-defined groups.

We therefore examined this correspondence in AI-READI [2], focusing on participants in the non-insulin-treated type 2 diabetes enrolment stratum. We first compared glucose outcomes across independently defined clinical profiles, then assessed whether CGM-derived groups recovered those profiles and their defining characteristics. Prediction analyses examined how much information CGM provided about clinical variables and profile membership, including its incremental value over HbA1c. Analyses across glycemic strata assessed whether the observed correspondence reflected within-stratum profiles or broader differences across the glycemic spectrum.

2 Methods

2.1 Cohort and CGM outcomes

We analysed AI-READI clinical data and ten-day CGM recordings sampled every five minutes [2]. Of 1,892 processed participants, 1,591 passed preprocessing: 564 healthy, 405 with prediabetes, 458 receiving oral or non-insulin injectable treatment for type 2 diabetes, and 164 receiving insulin. The non-insulin treatment stratum was the primary population.

We calculated mean glucose, time in range (70–180 mg/dL), and coefficient of variation over the first 48 hours and full record [3]. Full-record coefficient of variation was the prespecified primary outcome.

2.2 Clinical profiles

We clustered participants within each stratum using age, BMI, HbA1c, C-peptide, triglyceride-to-HDL ratio and waist-to-hip ratio. Preprocessing comprised iterative imputation, $\log(1 + x)$ transformation of HbA1c, C-peptide and triglyceride-to-HDL ratio, and standardization, fitted on training participants only. Healthy profiles used complete cases. We fitted k-means with 100 initializations on 325 training participants in the primary stratum and assigned the remaining 133

using frozen centroids. We used three profiles for each diabetes stratum and two for healthy and prediabetes participants.

2.3 Statistical analysis

Forward direction. We compared CGM outcomes across clinical profiles using Kruskal–Wallis tests and Dunn–Holm pairwise comparisons. We reported rank-based η^2 with 95% confidence intervals from 2,000 participant bootstrap resamples.

Reverse direction. We clustered CGM features comprising glucose summaries, time below 70 mg/dL, time above 180 and 250 mg/dL, mean amplitude of glycemic excursions, daily-profile components and autocorrelation summaries. Preprocessing and k-means were fitted on training participants, with frozen-centroid assignment for the remainder. We compared clinical variables across CGM groups using the same tests and assessed partition agreement using ARI [5], normalized mutual information and bias-corrected Cramér's V, with 1,000 label permutations. We predicted clinical variables using ridge regression and profile membership using logistic regression, comparing CGM alone, HbA1c alone and their combination. Linear regression predicted CGM outcomes from clinical variables with and without HbA1c. Evaluation used five-fold participant-level cross-validation repeated 20 times, stratified by profile, with preprocessing refitted within outer training folds. Prediction nulls used 200 permutations with fixed hyperparameters, increased to 1,000 for primary-analysis p values between 0.01 and 0.10. Interpretation criteria were prespecified.

Other strata and diagnosis. We repeated both directions within the other strata, matching CGM and clinical cluster counts. We assessed transfer of the C2-associated variability pattern by assigning healthy and prediabetes participants to frozen type 2 diabetes centroids and comparing C2-assigned participants with the remainder, with Holm correction across strata. Pooled analyses assessed agreement with enrolment strata and combined stratum-profile labels, using within-stratum permutations to test additional profile correspondence. These analyses were prespecified; insulin-treated results were exploratory.

Sensitivity analyses. We examined training-only and complete-case subsets, assignment margins above 0.20, ANOVA, and two-group CGM clustering. Medication analyses were descriptive and stratified by GLP-1 or GIP/GLP-1 flags. In descriptive tables, counts of 1 to 4 participants were suppressed, with complementary suppression to prevent recovery.

Temporal reproducibility. To test reproducibility within the monitoring period, each CGM record was split into two non-overlapping windows (days 1 to 3 and days 4 to 6). CGM features, their transformations and k-means clustering ($k = 3$) were fit on first-window training participants and applied unchanged to both windows. We compared group assignments between windows in held-out participants and repeated the profile agreement in each window.

3 Results

3.1 Clinical profiles differ in glucose level and variability

Clinical profiles separated glucose level and variability differently. C3, which had the highest HbA1c, C-peptide and triglyceride-to-HDL ratio, had higher mean glucose and lower time in range. C2, characterized mainly by younger age and higher BMI, instead had lower coefficient of variation (Table 1; Fig. 1). All three outcomes differed across profiles at both windows (all $p < 0.0001$), but distributions overlapped substantially. Full-record effect sizes ranged from 0.052 to 0.078 (Table 2).

Variability did not follow the same profile ordering as glucose level. Despite their differences in C-peptide and triglyceride-to-HDL ratio, C1 and C3 showed little separation in coefficient of

Table 1: Baseline characteristics by clinical profile in non-insulin type 2 diabetes. Continuous variables are median [IQR], categorical variables n (%).

	C1	C2	C3	All	P
n	192	111	155	458	
Age, years	64 [56–70]	51 [47–58]	68 [63–76]	63 [54–70]	<0.001
Female sex	114 (59.4)	85 (76.6)	48 (31.0)	247 (53.9)	<0.001
Race					<0.001
White	80 (41.7)	43 (38.7)	90 (58.1)	213 (46.5)	
Black or African American	55 (28.6)	45 (40.5)	22 (14.2)	122 (26.6)	
Asian	52 (27.1)	11 (9.9)	29 (18.7)	92 (20.1)	
Other or not reported	5 (2.6)	12 (10.8)	14 (9.0)	31 (6.8)	
BMI, kg/m^2	26.8 [24.0–29.7]	38.7 [35.7–42.8]	29.1 [26.2–31.9]	29.6 [25.9–35.0]	<0.001
HbA1c, %	6.2 [5.7–6.6]	6.2 [5.8–6.7]	6.7 [6.2–7.3]	6.3 [5.8–6.9]	<0.001
C-peptide, ng/mL	2.3 [1.7–3.1]	3.3 [2.4–4.9]	4.4 [3.2–6.0]	3.1 [2.2–4.4]	<0.001
Triglyceride-to-HDL ratio	1.8 [1.3–2.5]	2.8 [1.9–3.9]	4.5 [3.1–6.7]	2.7 [1.7–4.2]	<0.001
Waist-to-hip ratio	0.91 [0.86–0.96]	0.91 [0.87–0.97]	1.00 [0.95–1.05]	0.94 [0.89–0.99]	<0.001
Metformin	140 (72.9)	62 (55.9)	109 (70.3)	311 (67.9)	0.007
GLP-1 or GIP/GLP-1 agonist	62 (32.3)	63 (56.8)	53 (34.2)	178 (38.9)	<0.001
SGLT2 inhibitor	33 (17.2)	16 (14.4)	50 (32.3)	99 (21.6)	<0.001
Sulfonylurea	13 (6.8)	5 (4.5)	30 (19.4)	48 (10.5)	<0.001

Triglyceride-to-HDL ratio computed from mg/dL values. Missing values: HbA1c 30, C-peptide 25, triglyceride-to-HDL ratio 23, waist-to-hip ratio 1. P values compare profiles (Kruskal-Wallis or chi-square test) and are descriptive, without correction for multiple comparisons. Race categories with fewer than 5 participants in any profile are combined. Site, blood pressure, LDL cholesterol, other medications, CGM coverage and the data split are in Supplementary Table **??**.

variation at 48 hours (difference -0.76 percentage points, 95% CI $[-2.06, 0.63]$) and only a small difference over the full record (-1.37 $[-2.49, -0.21]$). C2 had lower variability than both profiles. At 48 hours, its coefficient of variation was 2.26 percentage points lower than C1 (95% CI $[0.99, 3.56]$) and 3.02 points lower than C3. The same ordering persisted over the full record.

Table 2: CGM differences across clinical profiles in the non-insulin type 2 diabetes enrolment stratum ($n = 458$). Rank-based η^2 with 95% participant bootstrap confidence intervals. All omnibus comparisons had $p < 0.0001$. Pairwise patterns were supported by Dunn tests with Holm correction.

Outcome	First 48 hours	Full record	Between-profile pattern
Mean glucose	0.068 [0.029, 0.120]	0.057 [0.021, 0.109]	Higher in C3 than C1 and C2
Time in range	0.064 [0.025, 0.116]	0.078 [0.038, 0.133]	Lower in C3 than C1 and C2
Coefficient of variation	0.042 [0.015, 0.087]	0.052 [0.023, 0.100]	Lower in C2 than C1 and C3

The variability association was less consistent across participant subsets than the glucose-level associations. Full-record η^2 was 0.021 among the 325 training participants and 0.139 among the 133 held-out participants. The corresponding values at 48 hours were 0.013 and 0.111. Random-split comparisons provided limited evidence of an unusually large gap ($p = 0.043$ for the full record and $p = 0.084$ at 48 hours). Mean glucose and time in range showed no comparable split dependence.

Medication patterns also differed across profiles. GLP-1 or GIP/GLP-1 flags were more frequent in C2 (57%) than in C1 (32%) or C3 (34%). The full-record C2–C1 variability contrast was -2.77 percentage points (95% CI $[-4.46, -1.05]$) among participants without these flags and 0.51 ($[-0.99, 2.02]$) among flagged participants. These descriptive comparisons did not establish a

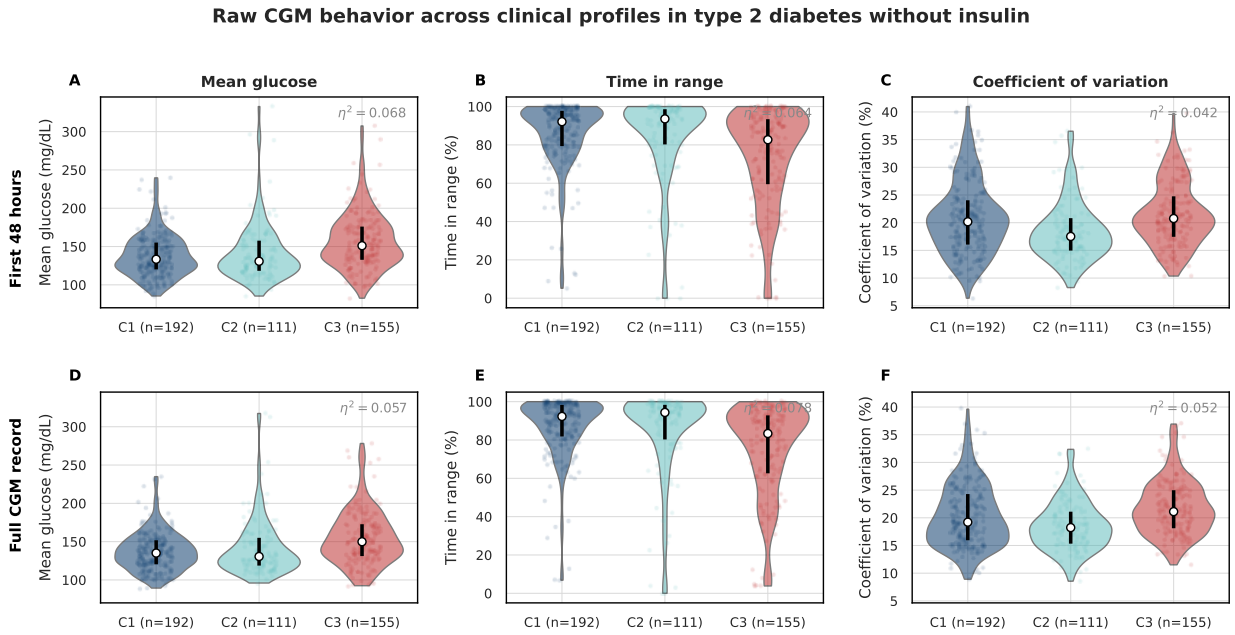


Figure 1: CGM outcomes across clinical profiles in the non-insulin type 2 diabetes enrolment stratum. Panels A–C show the first 48 hours and D–F the full record. Violins show participant distributions, with individual observations, medians and interquartile ranges. Annotated effect sizes are rank-based η^2.

treatment interaction or explain the profile association.

3.2 Low cluster agreement coexists with predictive clinical information

CGM clustering identified three groups of 188, 182 and 88 participants, with progressively higher glucose levels and variability (Fig. 2). Median mean glucose increased from 120 to 147 to 185 mg/dL, while median coefficient of variation increased from 16.4% to 21.1% to 24.8%. The three-group solution matched the clinical profile count, although the silhouette criterion favoured two groups.

These CGM groups corresponded weakly to the clinical profiles. Agreement was above the permutation null but small (ARI 0.039, 95% CI $[0.020, 0.068]$, $p = 0.001$; bias-corrected Cramér's V 0.205). C3 was enriched in the highest-glucose group and C2 in the lowest-glucose group, but each CGM group contained participants from all three clinical profiles.

Clinical differences across CGM groups were concentrated in HbA1c (η^2 0.422), with smaller effects for the other variables (η^2 0.002–0.061; Table 3). Prediction analyses supported this pattern. CGM features predicted HbA1c with an out-of-sample $R^2 = 0.590$ (95% CI $[0.455, 0.682]$), whereas prediction was weaker for age ($R^2 = 0.104$), BMI (0.062) and waist-to-hip ratio (0.050), and negligible for C-peptide (0.000) and triglyceride-to-HDL ratio (0.002).

In the opposite direction, the six clinical variables predicted full-record mean glucose and time in range with R^2 of approximately 0.51. Removing HbA1c reduced these values to 0.02 and 0.04, respectively. Prediction of coefficient of variation was weaker and declined less, from 0.079 to 0.051. Thus, predictive correspondence with glucose level depended strongly on HbA1c, while the remaining variables retained limited predictive information about variability.

Despite low partition agreement, CGM features retained information about clinical profile membership. Classification achieved a macro AUROC of 0.649 (95% CI $[0.612, 0.686]$) and balanced accuracy of 0.448 ($[0.400, 0.496]$), compared with a chance balanced accuracy of 0.333 (permutation $p = 0.005$ for both metrics). Adding CGM features to HbA1c increased AUROC from 0.605 to 0.660, a gain of 0.055 ($[0.017, 0.094]$). CGM therefore provided predictive clinical information without closely reproducing the clinical partition.

The prespecified criteria classified the reverse analysis as asymmetric because the age effect narrowly exceeded its threshold. This classification held in only 3 of 15 partition and participant-subset combinations. The criteria for reciprocal weakness were not met, and profile classification fell just below the prespecified AUROC threshold of 0.65 despite exceeding chance. These classifications were sensitive to the cutoffs; the estimates supported low partition agreement with modest residual clinical information.

Table 3: Effect sizes in both directions in the non-insulin type 2 diabetes enrolment stratum. Forward comparisons assess full-record CGM outcomes across clinical profiles. Reverse comparisons assess clinical variables across CGM groups derived from extended full-record features ($k = 3$). Values are rank-based η^2 with 95% participant bootstrap confidence intervals.

Direction	Variable	η^2 [95% CI]
Forward	Mean glucose	0.057 [0.021, 0.109]
	Time in range	0.078 [0.038, 0.133]
	Coefficient of variation	0.052 [0.023, 0.100]
Reverse	Age	0.061 [0.024, 0.113]
	BMI	0.020 [0.001, 0.060]
	C-peptide	0.002 [0.000, 0.029]
	Triglyceride-to-HDL ratio	0.025 [0.004, 0.069]
	Waist-to-hip ratio	0.028 [0.005, 0.069]
	HbA1c	0.422 [0.343, 0.497]

3.3 Correspondence varies across glycemic strata

Clinical profiles in healthy and prediabetes participants differed mainly in mean glucose, with full-record η^2 of 0.021 (95% CI $[0.003, 0.054]$) and 0.071 ($[0.028, 0.124]$), respectively. Differences in coefficient of variation were negligible ($\eta^2 \leq 0.002$). In insulin-treated participants, differences were small and less consistent across windows (exploratory; Fig. 3A).

We next examined whether the C2-associated variability contrast extended beyond type 2 diabetes using the fixed type 2 diabetes centroids. This assigned 88 healthy and 89 prediabetes participants to C2. The full-record coefficient-of-variation difference between C2-assigned participants and the remainder was +0.17 percentage points (95% CI $[-0.77, 1.12]$) in healthy participants, -1.04 ($[-1.88, -0.12]$) in prediabetes and -2.39 ($[-3.41, -1.43]$) in type 2 diabetes. The prediabetes comparison did not meet the significance threshold after correction across strata (Holm-adjusted $p = 0.065$). Point estimates suggested a smaller contrast outside type 2 diabetes, but these comparisons did not establish between-stratum heterogeneity. C2 assignment was not associated with detectable differences in mean glucose or time in range in healthy or prediabetes participants.

Low agreement between CGM groups and independently derived clinical profiles persisted across the other strata (ARI 0.004–0.013). Nevertheless, CGM features predicted profile membership in

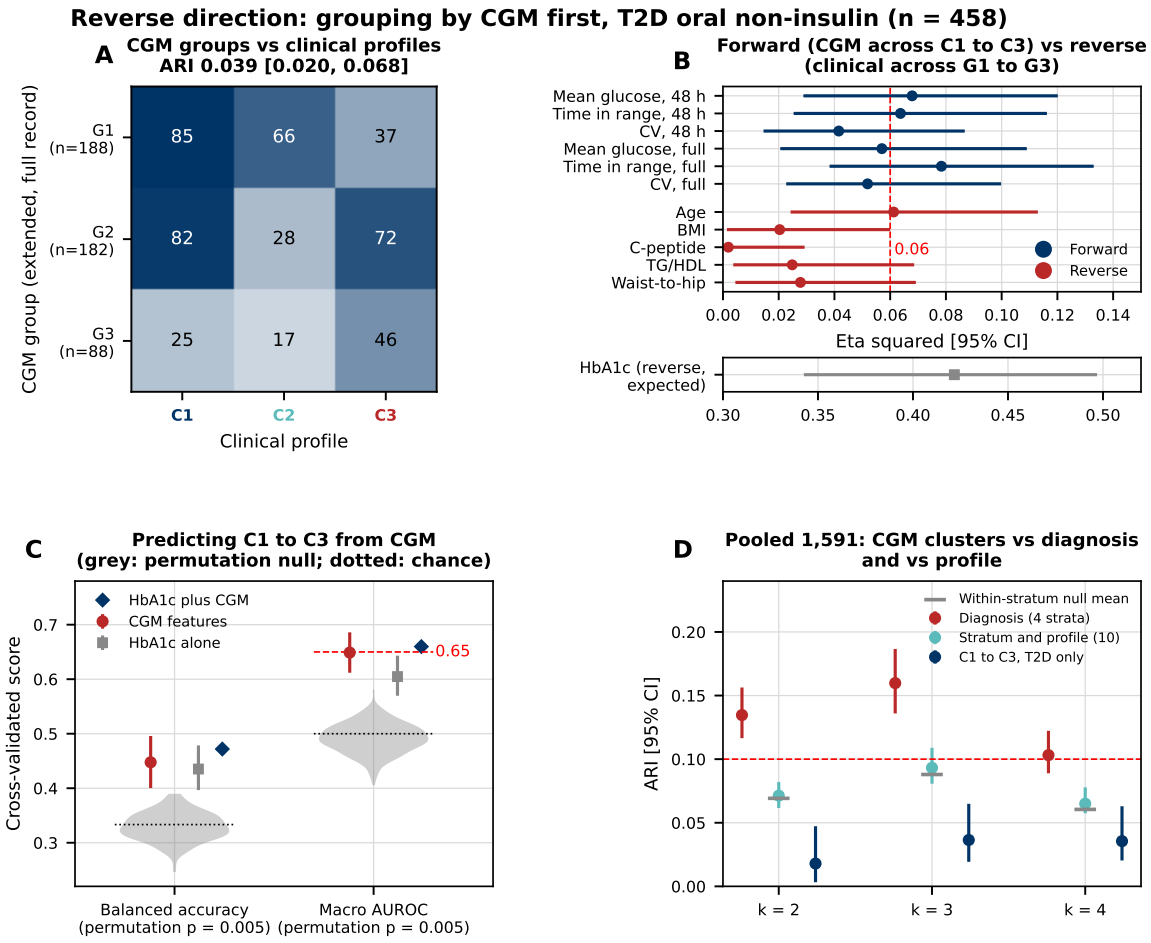


Figure 2: Agreement and prediction between CGM groups and clinical profiles. (A) Participant counts by CGM group and clinical profile. (B) Forward and reverse effect sizes with 95% confidence intervals; HbA1c uses a separate scale. (C) Profile classification from CGM, HbA1c and their combination, with permutation nulls and chance references. (D) Agreement of pooled CGM groups with enrolment strata, combined stratum-profile labels and clinical profiles within the primary type 2 diabetes stratum. The within-stratum permutation reference preserves stratum membership.

healthy and prediabetes participants, with AUROCs of 0.685 and 0.741, respectively, compared with 0.664 and 0.628 using HbA1c alone. Thus, weak agreement between unsupervised partitions coexisted with predictive clinical information outside type 2 diabetes.

Across CGM groups, differences in clinical variables other than HbA1c were generally small. An exploratory exception was C-peptide in insulin-treated participants, with full-record η^2 of 0.085 (95% CI $[0.019, 0.197]$), compared with 0.021 at 48 hours. This window dependence limited interpretation of the association as a consistent feature of the insulin-treated stratum.

In the pooled cohort, CGM groups were associated with enrolment stratum (ARI 0.160, 95% CI $[0.136, 0.187]$, $p = 0.001$; Fig. 2D). Restricting these pooled assignments to the primary type 2 diabetes stratum yielded similarly low agreement with C1–C3 as clustering that stratum separately (ARI 0.037 versus 0.039). Agreement with combined stratum-profile labels was 0.093, compared with a mean of 0.088 under within-stratum permutations ($p = 0.001$). Clinical profiles therefore

contributed a small additional association beyond enrolment stratum.

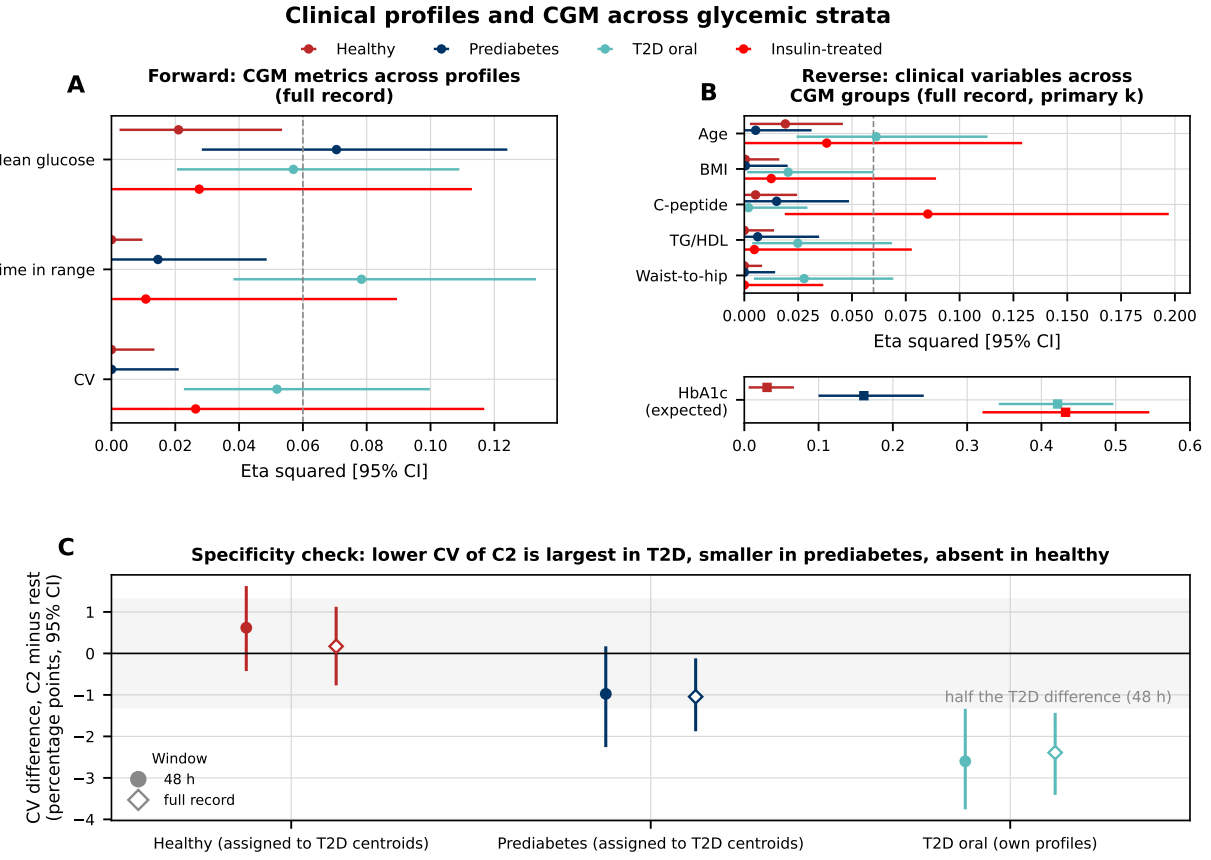


Figure 3: Clinical–CGM correspondence across glycemic strata. (A) Full-record CGM differences across each stratum's independently derived clinical profiles. (B) Clinical differences across CGM groups, with HbA1c shown on a separate scale. (C) Coefficient-of-variation differences between C2-assigned participants and the remainder. Healthy and prediabetes participants were assigned using the fixed type 2 diabetes centroids. Negative values indicate lower variability among C2-assigned participants. Points and intervals show estimates and 95% participant bootstrap confidence intervals. Insulin-treated analyses were exploratory.

4 Discussion

4.1 Principal findings

Clinical profiles distinguished glucose level and variability differently, but neither yielded clearly separated CGM distributions. Low agreement with CGM-derived groups coexisted with predictive information about profile membership beyond HbA1c. This pattern, also observed in healthy and prediabetes participants, indicates that clinical information can be present in CGM without determining its dominant clustering structure. Because CGM groups were moderately reproducible across non-overlapping windows while their agreement with the profiles stayed low in each window, the weak correspondence is not accounted for by unstable CGM groupings.

4.2 Relation to previous work

Previous studies identified clinically relevant diabetes subgroups, although continuous variables sometimes predicted progression and treatment response as well as or better than cluster labels [1, 4]. Our findings extend this question to free-living glucose, showing that differences between clinical profiles do not necessarily translate into corresponding CGM groups.

HbA1c accounted for much of the predictive association with glucose level. In contrast, C-peptide and triglyceride-to-HDL ratio were poorly predicted, and profiles differing in these markers showed little separation in variability. The lower variability associated with C2 also varied descriptively by medication context and glycemic stratum. These observations support interpreting the profiles within the treatment and population conditions studied, rather than as fixed glucose phenotypes.

4.3 Limitations

The profiles were cohort-specific and differed from established diabetes subtype classifications. HbA1c contributed to their construction, limiting the independence of glucose-level outcomes as validation measures. The primary variability outcome was sensitive to participant subset, and incomplete medication information prevented assessment of treatment contributions.

Correspondence depends on feature selection, cluster count and clustering stability. Correlated CGM summaries may emphasize glucose level, while omitting relevant temporal information. Insulin-treated analyses were exploratory. The temporal checks used 3-day windows within a single monitoring period of about ten days, so they address reproducibility within that period rather than long-term stability.

Eligibility criteria inherited from a forecasting pipeline included the availability of wearable-data, potentially limiting generalization. External validation is needed and the findings do not establish utility for treatment selection or clinical outcomes.

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